

Views from chief veterinary officers about decision making during animal health emergencies: A qualitative analysis

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ABSTRACT

This study's objectives were to investigate how delegates of the World Organisation for Animal Health (WOAH), particularly chief veterinary officers (CVOs), make decisions requiring immediate action, having national/international impacts on animal health and welfare and/or human health, and how the process creates new policies and practices. We interviewed 33 CVOs from 6 continents, on their background, jurisdiction, institutional structures, and decision-making processes, including types of decisions made, who they consulted, information used (and wished they knew), political, public and other influences, and resultant policy changes. The CVOs also discussed surprising and challenging phenomena, and what they learned. Qualitative analysis was conducted on interview transcriptions. Most CVOs had similar decision-making processes. They followed established protocols and national legislation aligned with WOAH international standards, relying on multidisciplinary teams of experts in science, economics, policy, and law, and those with knowledge of local field conditions, for guidance. Insufficient information and conflict between scientific evidence and political/economic pressure were common themes. Although stressful, most CVOs were committed to their work and felt they made valuable contributions towards both animal and human health. The findings suggest that regardless of background or specific geographical context, CVOs follow established protocols and need to have the ability to make informed subjective judgements as part of their decision making. Thus, CVO qualifications include subject matter knowledge and specific leadership qualities, which need to be considered when making CVO appointments. Presently, veterinarians receive inadequate training on integrating evidence and other factors to make informed, 'good' decisions. This study's findings should be considered when developing the educational programs for veterinary students and established practitioners.

1. Introduction

1.1. Background

National veterinary authorities (VAs) operate at the intersection of animal, human, and environmental health, reflecting the interconnectedness of One Health systems. This unique positioning gives VAs a pivotal role in managing animal health and welfare, zoonoses,

foodborne risks, and antimicrobial resistance. In many countries, particularly those with intensive animal production systems, the economic stakes in VA decision making are considerable. Disease outbreaks can severely disrupt livestock sectors, affecting food supply chains, international trade, and rural livelihoods (Grace, 2015; Otte et al., 2004). However, beyond the biological threat posed by disease, the decisions made in response—such as movement bans, culling policies, or import restrictions—also carry significant economic consequences. These

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policy actions can redistribute losses across sectors and stakeholders, influencing who bears the economic burden of the crisis. As a result, VA decisions, while supported by the science-based recommendations of [World Organisation for Animal Health \(WOAH\)](#) international standards, often become politically charged, attracting lobbying from affected producer groups, industry associations, and even consumer interests seeking to protect their economic positions ([Zinsstag et al., 2011](#)). This dual impact—where both disease and governance responses shape economic outcomes—underscores the complexity of decision making in VAs, particularly in emergency contexts where time and information are constrained. It is therefore important to better understand the science-policy interface when interacting with WOAH delegates and other VAs. How evidence is used and how policy is communicated requires understanding of the end users (i.e., decision makers).

1.2. Veterinary roles in decision making

Veterinarians play a crucial role in decision making across various levels, from clinical diagnoses and treatment of individual animals to the management of animal health and welfare at regional, national, and international scales. Every decision they make is influenced by a complex interplay of ecological, social, and political factors, which adds layers of uncertainty to the process. The inherent complexity of veterinary decision making means that even routine emergencies require careful planning, while true crises—characterized by unpredictability, resource constraints, and public scrutiny—demand adaptive leadership ([Glanville and Christie, 2011](#)).

Chief Veterinary Officers (CVOs), who serve as national authorities on animal health, must navigate these challenges while balancing science, policy, and public expectations. The role of a CVO is multifaceted, requiring not only technical expertise but also leadership, strategic thinking, and diplomacy. Insights from past and present CVOs highlight the breadth of their responsibilities, from shaping national biosecurity strategies ([Hueston, 2003](#); [Gibbens, 2015](#); [Foddai et al., 2020](#)) to engaging with international organizations such as the WOAH ([Paskin, 2018](#)). Former CVOs, such as Preben Willeberg, Denmark's CVO (1999–2007), have emphasized the importance of communication, collaboration, and ethical decision making, especially under conditions of limited data—where prior experience with similar situations is often essential ([Willeberg, 2012](#); [Lees, 2014](#)). The increasing interconnectedness of animal, environmental and human health further reinforces the need for a One Health approach, integrating veterinary science with broader public health and environmental considerations, a view also advocated by Harpreet Kochhar, former Canada CVO ([The Crest, 2021](#)).

1.3. Scientific evidence vs. politics

Scientific evidence is widely recognized as essential to sound decision making, yet politics and public opinion often play a significant role in shaping animal health policies. The mission of WOAH is to improve animal health and welfare worldwide, thereby ensuring a better future for all. The WOAH standards ([World Health Organization \(WHO\), 2002](#)) play a crucial role in supporting science-based decision making by providing internationally recognized recommendations to ensure safe trade, public health, and economic growth. These standards support policies and actions to ensure they are grounded in the latest scientific evidence, promoting transparency and consistency across the globe to effectively address global animal health challenges. The World Organization for Animal Health has developed a science system that describes how WOAH accesses the best available science to support its core activities ([World Organisation for Animal Health WOAH, 2023](#)). While WOAH standards and other scientific evidence provide a foundation for policy formulation, competing interests—such as economic considerations, lobbying groups, and social attitudes—can sometimes overshadow evidence-based approaches ([World Health Organization \(WHO\), 2002](#); [Pfeiffer, 2006b](#); [Foddai et al., 2020](#)). A World Health

Organization study group examined veterinary public health programs and needs throughout the world; two of their conclusions were that new approaches were needed to obtain and manage reliable data, and that veterinary public health programs had to be flexible and adaptable, and involve all stakeholders ([WHO, 2002](#)). [Pfeiffer \(2006b\)](#) discussed the challenges that stakeholder risk perception and dwindling public trust in scientific evidence pose for development of science-based disease control policies. [Foddai et al. \(2020\)](#) emphasized the importance of surveillance as evidence to support decision making. Policymakers must balance the urgency of action with the need for robust scientific validation, a challenge exemplified in historical cases such as Portugal's bovine spongiform encephalopathy (BSE) crisis ([Gonçalves, 2000b](#)) and the ongoing debate over badger culling for bovine tuberculosis (TB) control in the UK ([McCulloch and Reiss, 2017](#)). In such instances, public trust in VAs can be jeopardized if political motivations appear to take precedence over scientific integrity. [Gonçalves \(2000b\)](#) stated that building trust between scientists and policy makers is critical in order to bridge the gap between science and politics; however, the political system may be unable (or unwilling) to change its ways. In addressing the bovine TB-badger issue in the UK, [McCulloch and Reiss \(2017\)](#) recommended conducting an animal welfare impact assessment to obtain objective data on policy options and then having independent experts analyze the data within a moral framework; such activities would be helpful to decision makers.

Effective decision making relies on the availability of high-quality, timely, and relevant information. While evidence-based decision making is an aspirational standard, in practice, veterinary and public health authorities must often make critical decisions under conditions of incomplete or evolving data ([Brownson et al., 2009](#)). This tension is especially evident during emergencies, where time-sensitive choices must be made in the absence of perfect information. The concept of evidence-informed decision making in public health emphasizes the use of the best available evidence from various sources to inform and improve animal and public health practice. This approach requires decision-makers to integrate multiple factors into their conclusions and recommendations, even when faced with uncertainty.

Moreover, the landscape of information is not always neutral. [Oreskes and Conway \(2010\)](#), in 'Merchants of Doubt', have shown how scientific uncertainty can be deliberately amplified by interest groups to delay or delegitimize regulatory action. Similar dynamics can occur in animal health, where actors with vested economic interests may challenge or minimize the implications of emerging data to avoid or delay the implementation of disruptive control measures. One example of this is the UK bovine TB-badger controversy ([McCulloch and Reiss, 2017](#)). Badger culling was implemented in some areas to decrease bovine TB; however, it did not always lead to lower cattle infections. A political divide emerged, with Conservative politicians generally in favor of culling as a necessary evil and Labour against, with the latter criticizing the scientific evidence used in the policy. Furthermore, opponents of culling questioned the high cost of culling each badger, given the state of the economy. In such cases, information becomes a contested resource—both a technical input and a political tool—used to shape narratives, justify decisions, or forestall interventions. These dynamics further complicate the role of CVOs (who have different levels of decision-making authority, depending on the structure of their organization), and indeed anyone involved in the process, who must evaluate not only the strength of the evidence but also its political and strategic implications in real time, as part of the entire decision making process. This is of course an integral part in any high-level decision making process, not only those involving animal health emergencies.

Tools and conceptual frameworks, such as info-gap decision theory, have been developed to support decision making under uncertainty. These approaches help policymakers make robust decisions despite incomplete data, by focusing on the range of acceptable outcomes and prioritizing flexibility over precision. However, such tools must be used alongside institutional judgment and governance structures that define

what types of decisions are permissible under uncertainty.

1.4. Governance and structure

Despite the existence of WOA international standards, governance structures for veterinary decision making vary significantly across countries. In the European Union, CVOs play a central role in shaping disease control regulations and benefit from a harmonized legal framework that supports precautionary measures (Mourits et al., 2010; Stärk et al., 2022).

The precautionary principle is a strategy that involves taking preventive action in the face of uncertainty to protect animal health and welfare, e.g., even if evidence is lacking for some cause-and-effect relationships. In the context of the Sanitary and Phytosanitary (SPS) agreement (World Trade Organization (WTO), 2025), the precautionary principle allows the adoption of provisional measures when there is insufficient scientific evidence about potential risks to human, animal, or environmental health. These measures must be based on available pertinent information, and countries are required to seek additional information to clarify the scientific uncertainty and review the measures within a reasonable period. This principle is institutionalized in many countries such as European Union member states (European Commission, 2000); others like the United States often emphasize risk-benefit analysis, requiring higher thresholds of evidence before interventions can be triggered (Vogel, 2012). These jurisdictional differences reflect broader cultural orientations towards uncertainty, scientific authority, and the role of the government in risk management (Jasanoff, 2005).

In some countries, governance is decentralized, relying on inter-governmental collaboration across local, state, and federal levels (Glanville and Christie, 2011; Animal Health Australia, 2015; Woods et al., 2019). Regardless of structure, successful decision making in veterinary public health depends on coordinated efforts among government agencies, scientific experts, and industry stakeholders (World Health Organization (WHO), 2002; Reynolds and Donaldson, 2005). International trade, public health, and wildlife conservation add further layers of complexity, requiring CVOs to balance local and national interests with compliance to global standards and expectations (Hueston, 2003; McCulloch and Reiss, 2017; Department for Environment Food and Rural Affairs, Animal and Plant Health Agency, 2022). These decisions are shaped not only by scientific data but also by regulatory local, national, and international frameworks, institutional cultures, and political feasibility. As such, the interplay among information, governance, and context is critical to understanding how CVOs navigate decision making in animal health, particularly under emergency conditions.

1.5. Study objectives

This study investigates how CVOs and WOA delegates navigate high-stakes decision making in complex multisectoral contexts, where animal, human, and environmental health intersect. Specifically, we aim to explore (1) how animal health related decisions are made under conditions of uncertainty and incomplete information, particularly when economic and political pressures are significant; and (2) how these decision-making processes are shaped by, and in turn influence, public policies, governance frameworks, and international, national, and local regulatory practices.

By drawing on first-hand perspectives of CVOs and WOA Delegates, this research contributes novel empirical insights into the underexplored space among science, policy, and strategic action in the context of animal health. The findings are not only of academic relevance, but also hold practical and educational value for current and future veterinary professionals navigating complex One Health governance challenges.

2. Materials and methods

2.1. Study population

All CVOs and WOA Delegates (with a DVM), both currently employed and ex-officers, were eligible for inclusion. The authors first used non-probability, purposive sampling to identify interviewees, as is aligned with sampling practices for in-depth qualitative interviews (Bernard, 2017). Potential interviewees, in their role as WOA delegates, were initially identified by Author 4. The interviewees were not selected randomly, but rather chosen purposively, with a focus on CVO and WOA Delegate decision makers at various levels of authority (e.g., local, provincial, national), across world regions, and who were either personally known to the investigators or recommended by other interviewees using snowball sampling (Bernard, 2017).

In qualitative research, the sampling scheme is not to generalize to a population as it would be with probability sampling, but rather is to describe a phenomenon of interest (Wutich and Bernard, 2023). Snowball sampling is a relevant strategy when participants are difficult to reach, including if there are a “few members who are scattered over a large area” or “they are members of an elite group...and don’t care about your need for data” (Bernard, 2017). Thus, it was an appropriate sampling strategy for this study.

The invitation letter to the study is provided in Appendix A.

2.2. Privacy and consent

Our protocol was approved by our institution’s Review Board. No compensation was offered to participants. Informed consent was obtained; all responses were kept confidential.

Each interview took place at a time and place that was convenient for the interviewee and assured privacy. Informed consent was obtained before interviews began; the consent form is provided in Appendix B. We refrained from asking sensitive questions such as those about identifiable information, or the “rightness” of decisions. Our primary focus was on understanding attitudes and perceptions, and the interviewees’ experience in decision-making situations. We did not ask questions that could lead to identification. Any materials with identifying information were either locked in a drawer or stored on our institution’s encrypted cloud storage, which requires a password to access. These identifiers were not linked to the interview recordings or transcriptions.

Personally identifiable information (name, e-mail address, telephone/fax number, video recordings) was collected, and stored securely until the manuscript was accepted for publication. All contact information was destroyed after interviews were completed and transcribed. Each investigator assigned a code which began with the investigator’s initials and was followed by a sequential interview number. The link between identifiers and code was maintained until publication of this work. The final results contain only general research themes, and do not include identifiable information about the participating veterinary officers.

2.3. Data collection (interviews)

Authors 1, 2, and 3 conducted confidential, recorded semi-structured Zoom interviews with CVOs from the 5 WOA regions (Africa, Americas, Asia and Pacific, Europe, and Middle East) between February 2023 and June 2024, to understand their attitudes and perceptions of the decision-making process, with a focus on animal health issues, and what their experience was in handling decision-making situations. All 3 interviewers hold both DVM and PhD degrees. All investigators have been involved in diverse aspects of veterinary research for many years.

Interviews were conducted in either English or French. Each was approximately one hour long. The interviews included both standard (yes/no or multiple-choice) and open-ended questions. They began with generic questions, such as the setting (demography, geographical

locations etc.) where the CVO worked, the role of veterinarians, other decision makers involved, available information (from the literature or the specific research done), how the CVOs reached their final decision (s), any follow-up plan(s) to modify the decision(s), and how ethical issues such as uncertainty (i.e., a lack of or limited knowledge) were addressed. The actual questions and probes are listed in [Appendix C](#). Interviews were conducted until no new ideas emerged.

2.4. Qualitative analysis

The Zoom interviews were recorded and a transcript was auto-generated. The transcripts were reviewed and checked for accuracy by the team in Microsoft Word in preparation for qualitative text analysis.

The interviews were uploaded into [ATLAS.ti 24](#) (Scientific Software Development GmbH, Berlin), a software program designed for text analysis and qualitative research. ATLAS.ti was used to code the data. We first conducted word analysis of the transcripts to derive codes and code groups. Additional codes were added inductively as new concepts emerged from the interviews. A set of 19 descriptive codes was developed (by Authors 1 and 6), based on topics that emerged from the interviews and based on guidelines in [Saldaña \(2009\)](#). These codes were used to characterize passages of interview text, and transcripts were re-reviewed as new codes emerged. All authors were in agreement regarding the final codebook. The codes were grouped into 6 categories ([Table 1](#)) and their relationship to one another was examined using ATLAS.ti. Then, codes and categories were used to identify key themes related to various aspects and factors in CVOs' decision-making process. The level of authority (none, low, moderate, high) of each CVO was also determined, based on their responses to interview questions.

While [ATLAS.ti 24](#) was used for most tasks in this study, the later version [ATLAS.ti 25](#) (Scientific Software Development GmbH, Berlin), was used to generate [Fig. 1](#) (Word Cloud).

Table 1

Codes developed from interviews with 33 chief veterinary officers (CVOs) and World Organization for Animal Health (WOAH) delegates from 5 WOAH regions. Each code was allocated to one of 6 higher-order code groups.

Code	Code Group
● Private practice, career progression ● Had great mentor, hindsight, experts, educators, who to consult ● Trust in country's animal health systems, communicate, be transparent with public, public acceptance ● Active interface with wildlife ● Advice to next generation ● Worked in USDA/government since vet school, never in practice, career progression ● CVO characteristics, interests, background, have technical knowledge, not be political appointee ● Improve reproductive/general herd health to help own/developing countries, international issues ● Country has large export market/trade issues: very dependent on it; economic concerns ● Country traceability issue ● Interactions, collaboration with industry, international counterparts, stakeholders	Beneficial attributes
● Jurisdiction, reporting structure ● Response to emergencies, situations ● New policies, practices, follow-up plans resulting from decision	International or other relations
● Decisions based on science and standards, not politics; decision-making process clear	Jurisdiction, structure
● Science evolves, new knowledge, technology	Responses to situations
● Competing priorities; nature of the job ● Pleasing the politicians, colleagues ● Concern for colleagues' wellbeing	Science-based decisions
	Competing concerns

3. Results

3.1. Codes, clouds and networks

The 19 codes developed and used to characterize text passages and identify themes in the interviews are shown in [Table 1](#). They are also grouped into 6 higher-order categories.

[Fig. 1](#) is an example of a Word Cloud produced by [Atlas.ti 25](#) software for the 33 interviews (interviewees' (CVOS') words only; i.e., the interviewers' (Authors 1, 2, and 3) words were excluded). In this example, the word threshold was set at 100, meaning that all words in the cloud appeared at least 100 times across the 33 interviews. The larger the size of the word, the more frequently it was mentioned by the interviewees. As might be expected, the word "decision" appeared most often. Other common words included "animal", "health", "people", "disease", "information", and "public".

[Fig. 2](#) is an example of a network produced by [Atlas.ti 24](#) software for the 33 interviews. It links the various codes in [Table 1](#), illustrating how they are related to each other, through relational properties (e.g. "is a result of", "may contribute to", etc.).

3.2. Findings from interviews

3.2.1. Background

The CVOs and WOAH delegates represented various countries in all 5 WOAH regions ([Table 2](#)). All 33 interviewees had various positions in government before being appointed CVO. Fifteen mentioned being in veterinary practice; 4 were never in practice and had only worked in governmental positions. Three had also been in academia, and 9 in food safety.

3.2.2. Jurisdiction

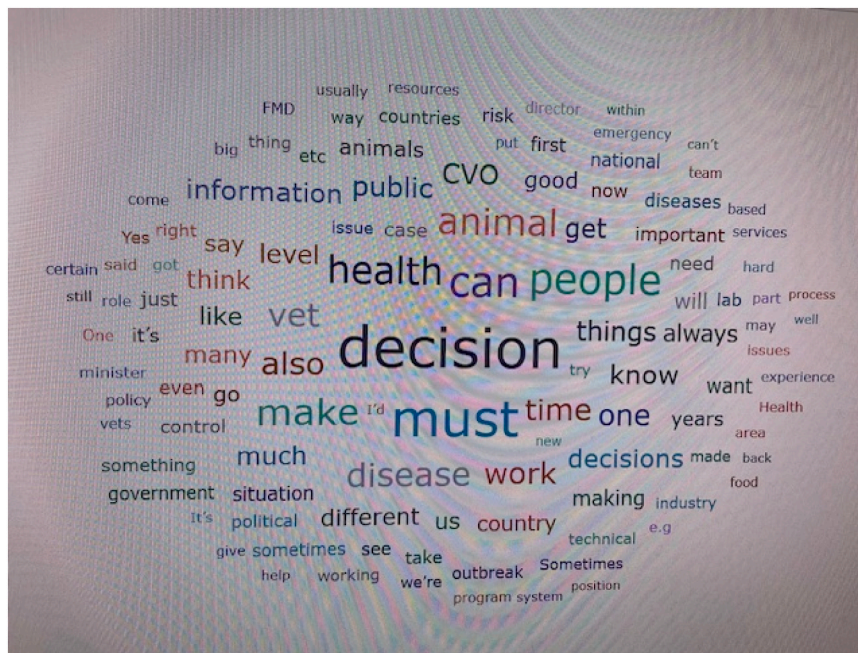
Most countries had states/provinces and districts (with headquarters and regional offices), each with its own CVO. Chief veterinary officers were responsible for varying numbers of staff. The majority of the national CVOs were WOAH Delegates.

The CVO reported to one or several figures, including ministers, secretaries, Director Generals, department heads, CEOs and administrators. The CVO was sometimes second in line, but more often was third or fourth in line. Chief veterinary officer reporting structures varied with the situation, e.g. the minister or secretary in an urgent situation, but a lower figure in more routine situations. Some CVOs advised their superiors, but were not necessarily the main decision makers. Although statistics were unavailable on how CVOs many also functioned specifically as crisis managers during a crisis, CVOs always had a role in the incident management system, as crisis managers in some form. Yet, all of them played significant roles in animal health management, including crisis management.

For 15 of the CVOs in this study, their position empowered their decision making to a large extent. For 7 CVOs, it was to some extent, and for others, it was only in certain areas (e.g. disease control but not policy development; or for routine work only). One CVO had no decision-making power; it was the minister's responsibility. Based on these responses, 15 CVOs were considered to have a high level of decision-making authority, 15 had a moderate level, 2 had a low level, and one had no authority. There did not appear to be a clear association between a CVO's level of authority and the experiences they had as CVO.

As mentioned above, CVOs always had some role in crisis management. Learning from the crisis was important, regardless of their level of decision-making authority:

"It's good to have these prepared documents, but when a crisis comes, you must take this prepared document and throw it into the basket; you'll find that the situation is totally different than in the papers. The most important issue is that you can learn." (CVO with high decision-making authority) and



The diagram is a conceptual map illustrating the relationships between 19 numbered boxes. The boxes are arranged in a circular fashion around a central box (9). The relationships are indicated by arrows with labels such as 'IS AN EXAMPLE OF', 'CONTRIBUTE TO', 'IS A RESULT OF', 'IS ASSOCIATED WITH', 'PLAYS A ROLE IN', 'MAY CONTRIBUTE TO', 'CONTRADICTS', and 'IS ASSOCIATED WITH'. The boxes contain various topics related to career progression, decision-making, and policy development in the field of animal health.

Box 9 (Central): Decisions based on science & standards, not politics-DM process clear

Box 1 (Top Left): 1) Private practice, career progression

Box 2 (Left): 2) Improve repro/ general herd health to help own/developing countries, international issues

Box 3 (Left): 3) Had great mentor, hindsight, experts, educators, who to consult

Box 4 (Top): 4) Jurisdiction, reporting structure

Box 5 (Top Right): 5) Response to emergencies, situations

Box 6 (Bottom Left): 6) Country has economic concerns, large export market/ trade issues: very dependent on it

Box 7 (Bottom Left): 7) Country traceability issue

Box 8 (Bottom Center): 8) Trust in country's animal health systems, communicate, be transparent with public, public acceptance

Box 9 (Central): Decisions based on science & standards, not politics-DM process clear

Box 10 (Top Right): 10) Interactions, collaboration with industry, international counterparts, stakeholders

Box 11 (Right): 11) Science evolves, new knowledge, technology

Box 12 (Bottom Right): 12) Competing priorities

Box 13 (Bottom Right): 13) Active interface with wildlife

Box 14 (Bottom Right): 14) Pleasing the politicians, colleagues

Box 15 (Bottom Center): 15) Advice to next generation

Box 16 (Top Left): 16) Worked in USDA/ government since vet school, career progression, never in practice

Box 17 (Right): 17) New policies, practices, follow-up plans resulting from decision

Box 18 (Top Right): 18) Concern for colleagues' wellbeing

Box 19 (Bottom Center): 19) CVO characteristics, interests, background, have tech knowledge, not be political appointee

Relationships:

- Box 1: IS AN EXAMPLE OF Box 9
- Box 2: IS AN EXAMPLE OF Box 9
- Box 3: IS AN EXAMPLE OF Box 9
- Box 4: IS ASSOCIATED WITH Box 9
- Box 5: IS ASSOCIATED WITH Box 9
- Box 6: IS AN EXAMPLE OF Box 9
- Box 7: IS AN EXAMPLE OF Box 9
- Box 8: IS ASSOCIATED WITH Box 9
- Box 10: IS ASSOCIATED WITH Box 9
- Box 11: IS ASSOCIATED WITH Box 9
- Box 12: IS ASSOCIATED WITH Box 9
- Box 13: IS ASSOCIATED WITH Box 9
- Box 14: IS ASSOCIATED WITH Box 9
- Box 15: IS ASSOCIATED WITH Box 9
- Box 16: IS AN EXAMPLE OF Box 9
- Box 17: IS ASSOCIATED WITH Box 9
- Box 18: IS ASSOCIATED WITH Box 9
- Box 19: IS ASSOCIATED WITH Box 9

Table 2
Number of chief veterinary officers (CVOs)/World Organization for Animal Health (WOAH) delegates interviewed from each WOAH region.

Region	Number of CVOs/delegates interviewed
Africa	3
Americas	8
Asia and Pacific	10
Europe	9
Middle East	3
Total	33

3.2.3. Responses to situations

Chief veterinary officers made decisions on a wide variety of issues. Typical decisions included monitoring diseases overseas, animal welfare controls, domestic disease eradication, farm restrictions, and funding decisions. Some unusual decisions mentioned involved BSE, the 2009

swine flu pandemic (influenza A/(H1N1)pdm09), SARS-CoV-2 infections and highly pathogenic avian influenza (HPAI) in minks, SARS-CoV-2 infections in hamsters and dogs, mpox (decisions involving both animal and human health), situations not seen before or for a long time or posing a big threat or in a new location, suspicion of a disease (positive lab result but no symptoms), and shipping issues with samples.

If an urgent situation needing a quick decision arose, typically protocols and legislation were followed. In general, although the CVO provided a technical decision based on scientific evidence, their superior was the final decision maker. The time between recognition of a problem and the decision made was generally short, e.g. < 1 day for labs to confirm/rule out a disease, although it could be longer due to complicating factors. The VA could then take action, such as declaring an outbreak to WOAAH, implementing control measures, or making an interim decision, within an hour.

To deal with the pressure of quick answering in situations of high uncertainty, CVOs trusted in people and experts and others' experience, having the best available scientific information, made interim decisions, and tried to be prepared.

If the CVO had never faced such a situation before or had only limited knowledge about it, they took various actions. These included consulting with experts, making interim decisions, assessing consequences of actions (or non-actions), seeking more context before acting, using legislation, giving transparent messaging, and listening to public opinion and stakeholders. The purpose of the expert consultation is to obtain an expert risk assessment, which could be done formally or as expert opinion. This information is used by the CVO to make a decision (e.g., on risk management). For example, WOAAH's Terrestrial Code Chapter 2.1 WOAAH describes the process for import risk assessment (World Organisation for Animal Health WOAAH, 2025).

One CVO, with a low level of decision-making authority, noted: *"The issue of scientific uncertainty and risk communication are intimately linked. In my office, in areas of scientific uncertainty, our messaging was transparent: This is what we know. This is what we don't know. These are the actions we're taking to bridge that gap. We'll keep you updated as either the science or our actions evolve, to justify why we're taking certain actions."* Another, who had no decision-making authority, preferred safer and practical options: *"If there's uncertainty, and the consequences are important for animal health or for public health, I'd advise the minister the safe option, choose the option that's good for animal or public health, and not for the economy...I'd rather focus on practical solutions, which of course, affect finances too; if you choose a practical solution, it's often cheaper too."*

3.2.4. Sources of information

For more assistance or information, CVOs consulted a variety of professionals: subject matter experts, colleagues, other CVOs, local vets/staff, superiors, public health professionals, university staff, communications people, and legal people. The most influential information included expert advice, science, legislation/protocol, ground verification of threat, historical context/different scenarios, university expertise, experience, and a central husbandry of farms and animals. It is crucial to consider a broad spectrum of information: *"We've always defined science broadly. It can be biological, political, economic, social science. But we talk about science-based decision making. So all of these elements are part of the risk assessment process... Be aware and prepared to inform and educate in response to pseudo science."* (CVO with low level of decision-making authority)

In making a decision, CVOs wished that they had had various kinds of information. These included more up-to-date information (to prevent being isolated from the real world), risk of farms being infected by wild birds, whether products from affected premises were traded (affects scale of measures to apply), reliable information about other countries connected with an incident, and information on wildlife.

3.2.5. Other influences

In an unusual situation, CVOs could make various alternative

decisions or choices, including less drastic measures, interim decisions, declaring an outbreak immediately or waiting if unsure (decisions were faster for major diseases that had more obvious signs or ramifications (e.g. economic or public health)), and doing nothing.

Time limitations or competing priorities or deadlines forced CVOs to take various actions. These included comparing alternative decisions (benefits/costs), recognizing that human health was the top priority, then animal welfare, ensuring that commodities were safe for trade according to WOAAH international standards, delegating, not ignoring routine activities in an emergency, taking the safe option for animal or public health (not necessarily the most economic one), starting the next steps at first suspicion, interim measures, and officially notifying WOAAH. One CVO, with a moderate level of decision-making authority, recommended focusing on manageable aspects: *"When there's a case that you can break the decision into smaller decisions to buy you time to get more information, try to do that."* Another CVO, who had a high level of authority, mentioned the different roles that they might take on: *"I often ask, am I here as the strategic epidemiologist, or as the CVO who looks across the animal health system in which I want to talk about exports or imports; I want to talk about welfare, or am I here as the chief technical officer? Those 3 roles are things that I need to define daily in why I'm being asked something. They are 3 different things."*

Financial or other resources constraints (e.g. lack of diagnostic equipment or tests) were an issue for most CVOs, although 13 said they had none or could readily obtain money from various sources. Such constraints resulted in limited staff, and other programs suffering in emergencies. One CVO said that each hour of delay was costly, but the right evidence was needed. Another had a cost-sharing agreement with industries for emergencies. Chief veterinary officers attempted to be creative and efficient in the face of constraints, and tried to balance restrictions vs. minimizing their economic impact.

Politics and power often played an (unwelcome) role in decision making. This experience did not appear to be associated with the CVO's own level of decision-making authority, i.e., it occurred regardless of how much authority (high, moderate, or low level) the CVO had. Two CVOs said that politicians (and wealthy farmers) controlled the money. Another reported that they gained credibility if they solved the problem to the politicians' liking. For many CVOs, although decisions were based on science, they could be political. Pressure from lobbyists was sometimes helpful. There was political interest if human health was also involved. Regulations were often politically decided. One CVO said their role was to keep people and politicians calm. Another said it was impossible for humans to make totally unbiased decisions; politics were part of them to some degree. Several CVOs said that politics were a factor, but the timing of announcing or taking action was more notable. The CVOs' technical advice was never questioned.

Public acceptance had a large impact on decisions. It influenced different types of decisions, including lab location, wildlife handling, depopulation and transportation methods, and animal welfare. Good communication (e.g., emphasizing that human health depends on animal health) and trust were key to public acceptance. Admitting that one doesn't know gave confidence and trust to people. One can be scientifically correct but morally/religiously wrong in the public's eyes.

When asked if they saw administrators guided by selfish rather than social interests, 18 CVOs said they never saw this; one said that people try to help each other and work together. Some administrators, however, were more interested in career progression (e.g. pleasing superiors) than public interest. Some administrators wanted a problem to increase their budget or staff or have a larger office, and to be more important. Some CVOs said that personal interests may influence, especially in animal welfare.

Generally, CVOs did not have to make decisions that may have compromised their careers, although 10 reported that they had. One said that in making tough decisions, they did not feel compromised, but knew it would upset some people. Another would resign before making an unethical decision.

Other influences on decision making were the industry and private sector, the mental health and welfare of people in the animal health system, and the differing perceptions of the general public and regulated public (e.g. consumers vs. producers): what is important may differ. Also, CVOs needed a good reason to inform the public to avoid causing confusion.

3.2.6. Things learned and/or implemented

Most CVOs had follow-up plans with different strategies to modify their decisions, if necessary. The decision-making process resulted in adjustments being made to policies, practices, protocols, control programs, emergency plans, laws, and compensation systems. In addition, there was more community engagement and farm surveillance. One CVO required everyone on their team to have their phone on them when responding to an incident, after experiencing a situation in which some team members were out of touch for several hours. Another CVO, with a low level of decision-making authority, mentioned the need to keep up with evolving conditions and thus protocol revision:

“Science evolves. Now we understand better where biological risk may have been, or what the true, economic, social or other scientific risk may have been, would we have done something differently? That entails going back: revisiting your protocol, adjusting, revising, updating, amending, or improving, based on new knowledge or knowledge gained from the response and the effectiveness of that response, not only how effective it was in managing the animal health or disease component, how effective was it in managing international trust, consumer, public and trading partner trust and industry support.”

The CVOs learned a variety of things as a result of going through the decision-making process. Teamwork was crucial and cited by many participants as a crucial element in the decision-making process. They reported that one can always learn and improve and make efforts not to repeat mistakes, and that CVOs must understand the system. They also reported the need for communication skills, honesty, trust, concern, cultural sensitivity, diplomacy, innovation, and people skills. Chief veterinary officers reported that their role could be stressful, but a feeling of agency helped them to feel less stressed; learning about their stresses helped. One CVO described feeling humble; they did an important job, impacting many animals and people. Another CVO highlighted the importance of taking time to make decisions. One CVO, who had a high level of decision-making authority, also noted the importance of prevention but lack of focus on prevention in implementation: *“Everyone likes fire engine trucks dousing the fire, but they don’t want to build fireproof houses. That is a frustrating aspect, to influence that policy of prevention... One of the big things I say: often it’s not lessons learned, but lessons not learned.”*

In decision making, the CVOs wished they had known or been trained during their training on a variety of topics. Two were in favor of a broad spectrum of education and interdisciplinary approach and One Health training: emergencies were not always confined to animal health or food safety. One CVO mentioned clinical practice, to help deal with the human-animal bond. Training in how to deal with people, behavior, and social aspects, and having patience was mentioned by several CVOs. Other topics included budget planning and economics, management, risk analysis, politics, policy, communications, industry, epidemiology, systems thinking, and training in effective decision making. One CVO said the best way to train was to work with others, especially in a crisis.

The CVOs were surprised by many things about the decision-making process. One mentioned that it was not the content, but rather their presence, charm, and understanding of the other person’s environment that impacted their ability to influence. Another was surprised by how much importance and autonomy the CVO had, and being alone at the end for the final decision. The continual change and need to keep adapting was also a surprise. Investments and consequences were often emotional for people, considering their love for animals. One CVO also noted surprise when their superiors disagreed with their decision, even

when it was correct. Another was surprised by how much special interest groups would go above the CVO, if they did not like the decision. Some CVOs also reported being surprised by the impact of politics on science, the potentially huge consequences of a decision for an entire industry or nation, and the complexity of the decision-making process.

Chief veterinary officers reported that many things about the decision-making process were challenging. These include waiting for more information and making decisions without enough information, and a willingness to accept a compromise solution that was scientifically valid but not the best one solution because it needed to be economically and politically viable. As one CVO with a moderate level of decision-making authority put it: *“The short answer is, you assume the worst until you have scientific evidence to the contrary, and make decisions on that worst case scenario. It’s easier to loosen restrictions than to tighten them afterwards... The challenge is not eradicating the industry before you eradicate the disease.”* Another mentioned the difference between target audiences: *“A key factor for us is perception of the general public vs. perception of the regulated public. When we work, say, with poultry producers, their perception of what’s most important, and how we should address a disease outbreak might differ from the consumer buying poultry. That’s probably our greatest challenge in decision making in many cases: balancing the perceptions in different groups.”* (CVO with moderate level of decision-making authority)

Other challenges included responsibility, obtaining resources, lack of transparency, promotion of prevention policy, conflicts, motivating people to reach decisions in non-crisis situations, having to consult others, doing things one never thought one would do, ensuring that all necessary aspects are taken into account, decisions involving human health, and politics.

3.2.7. Open remarks

The final interview question offered CVOs an opportunity to express any thoughts that were not covered in earlier questions. One participant shared that not all individuals could or should be CVOs; CVOs needed to be easygoing but not overly so, and not nervous. It is a difficult, time-consuming job, but keeps one focused and stimulated. Participants described that it was very satisfying to resolve an issue successfully. Political pressure can be overwhelming, but seeing the positive impact of good decisions was rewarding. One CVO advocated working with one’s strengths, focusing on what makes one happy and excited. They shared that CVOs need a vision of where one wants to be, and to look for opportunities. They also shared that personalities, relationships, and leadership styles impacted how decisions were made. A CVO must have technical qualifications, be able to communicate, and explain decisions, and should never be a political appointee. They also highlighted that not being an expert was OK, as having distance sometimes resulted in a clearer perspective. They explained that one cannot prepare for or learn everything on the job. Many CVOs commented on how stressful their position was, e.g.:

“It’s a challenging process because you must participate, i.e. consult others. You must listen to and understand others. As a top level leader, the position is hard. When you make your own decision, it’s easy. It’s only you. You collect information and make the decision. But sometimes I can’t make my own decisions. When I’m in this position I must consult others because their exposure and knowledge are different. So I must make collective decisions, not my own decisions.” (CVO with moderate level of decision-making authority)

Nevertheless, most CVOs found their work to be highly rewarding. One CVO, despite having a low level of decision-making authority, declared: *“I’m here to tell you I’ve got the best job in the world, and if I do this till the end of my career, I couldn’t be happier because I just loved what I did.”*

4. Discussion

This study provides new empirical insights into how CVOs navigate complex, high-stakes decisions on animal health and welfare issues. The findings reinforce the notion that CVOs operate at the nexus of science, policy, and politics, where technical knowledge must be rapidly applied in real-world contexts marked by uncertainty, institutional constraints, and socio-political pressures. To fully appreciate the findings of this comprehensive interview-based study on CVOs' decision-making process, one should be clear with the study objectives and methods used. First, our objective was not to determine the optimal way to handle animal health matters, but given established protocols and legislation in each country, to study how individual CVOs actually made their decisions and which factors influenced their decision-making process. Perhaps the major advantage in our study was to learn about challenges and surprising things the CVOs identified about the decision-making process.

In the remaining discussion, we will first address “*Features of the decision process and leaders*” (Section 4.1) and then discuss how CVOs navigate uncertainty and urgency (4.2). We then discuss the political economy of decision making (4.3), ‘*Trust, communication, and public accountability*’ (4.4), ‘*Implications for policy and education*’ (4.5), and what makes a “good” CVO (4.6). Finally, we note some of our study's limitations and strengths and topics for future research (4.7).

4.1. Features of the decision process and leaders

Making decisions is a challenging activity, although it is often done without thinking what goes into the process. Chief veterinary officers have particular responsibility to make well-informed decisions, as the outcomes can impact both animal and human lives. Chief veterinary officers must be able to interact effectively with a wide range of stakeholders, even those with opposing views, or who are misinformed about the situation.

McKenzie (2014) describes the “dual process theory” of decision making, which involves 2 systems. System 1, or “intuition”, is rapid, relatively effortless, and handles scarce information well. The decision-maker feels comfortable with the outcome. However, it may not perform well in uncertain situations. In contrast, more effort is put into System 2; it is more formal, and uses various methods, e.g., listing potential diagnoses, and considers uncertainty. It makes fewer errors than System 1, and can even correct System 1 errors. System 2, however, is more laborious, and relies heavily on good information.

When making a decision, one must account for details regarding the situation, one's team (extent of their involvement), and oneself (Perrin, 2021). The last includes one's values and decision-making style (acting on intuition vs. deliberating all aspects of the problem). Decision-making style can range from the leader taking full control and responsibility to delegating tasks to team members and allowing them to decide what to do (Perrin, 2021). However, it is likely that nobody operates at either of these extremes, except perhaps in a dire emergency, but rather, somewhere in the middle (Perrin, 2021).

Fischer et al. (2022) outline attributes that veterinary leaders should have. These include competence, integrity, adaptability, ability to cooperate, tact, and decisiveness. They also discuss various leadership styles, including authoritarian, permissive, bureaucratic, and democratic; each has advantages and disadvantages, depending on the situation. Veterinary leaders need to maintain contact with their stakeholders, delegate effectively, and during a crisis, must act decisively, even without sufficient information, or when the decision is unfavorable (Fischer et al., 2022). Indeed, this is exactly what our interviewees advocated.

4.2. Navigating uncertainty and urgency

A core challenge described by CVOs was making rapid decisions in

the face of limited or evolving information. Despite aspirations toward evidence-based decision making (Brownson et al., 2009), our findings show that many decisions must be made before a full scientific picture is available. This aligns with the premise of decision theory frameworks like info-gap theory, which emphasize robustness under conditions of deep uncertainty (Ben-Haim, 2006). Chief veterinary officers in our study addressed this challenge by relying on expert consultation, past experience, and interim measures, underscoring the adaptive and iterative nature of their leadership. The pragmatic reliance on partial knowledge further reflects the realpolitik of public health response (WHO, 2002; Pfeiffer, 2006a), especially where immediate action is required to mitigate potential animal and human health risks.

Mourits et al. (2010) state that when a contagious animal disease is of concern, the view of the final decision-maker is of particular interest. To this end, they discussed a multi-criteria decision-making tool for policy makers, using the example of controlling classical swine fever (CSF) epidemics in the EU. Twenty CVOs compared various control strategies for CSF epidemics, accounting for objectives concerning epidemiology, economics, and social ethics. The approach accounted for the context of the decision (e.g., its objective (here, control of CSF), stakeholders), and weighting of criteria used in comparing the strategies and the strategies themselves. In ranking various control options, the CVOs put the greatest weight on epidemiological criteria, followed by economic and social criteria. Similarly in the current study, our CVO participants emphasized scientific findings and economic factors in their decisions, as well as political factors. Social factors, however, were often less well-understood and thus not ranked as highly. In the Mourits et al. (2010) study, duration of the epidemic and number of infected herds were the most important epidemiological factors, farm losses and export restrictions were the most relevant economic factors, and efficacy and social-economic impact were ranked highest within social factors. The authors concluded that region-specific control within the EU was necessary, and that although this assessment technique was time-consuming and perhaps not useful during the stress of an actual outbreak, it was clearly useful in decision making regarding contagious diseases. The tool helped CVOs to organize their thinking around the complex and sometimes conflicting aspects of contagious diseases, and to arrive at a consensus with their colleagues about how best to handle the situation (Mourits et al., 2010).

Similarly, Kim et al. (2017) used multi-criteria decision analysis to study epidemiological, economic, and social-environmental criteria that 21 CVOs (or their representatives) in Asia and Oceania used in making decisions in evaluating control strategies for foot-and-mouth disease (FMD). Size of the outbreak area, cost of control measures, and farmers' mental health were important criteria. The economic criterion was more important in countries that had been free of FMD for at least 10 years. Overall, epidemiological factors ranked higher in the DM process than did economic or social-environmental factors, similarly to the Mourits et al. (2010) study.

Humblet et al. (2012) surveyed 40 experts, including CVOs, to rank 100 diseases in Europe, based on their severity, morbidity, mortality, and economic impact. The results demonstrated that in making decisions, one must consider the severity of the disease and its consequences to all species involved (the animal(s) themselves and the people affected, and the costs and benefits of containment and mitigation (Humblet et al., 2012). Of the 100 diseases evaluated, the 5 considered of greatest importance were Nipah virus encephalitis, Venezuelan equine encephalomyelitis, highly pathogenic avian influenza, West Nile fever, and botulism, all of which are both zoonotic and common. Humblet et al. (2012) state that more attention is now being paid to prioritizing diseases, based on prevention strategies, particularly those of significance to public health. Similarly in the current study, CVOs prioritized protecting public health in their decision making.

Von Dobschuetz et al. (2015) surveyed CVOs in 94 countries on their surveillance systems for zoonotic influenza viruses. They found that some countries, concerned with viruses capable of inducing a pandemic,

looked for them only in pigs. The authors recommended expanding surveillance systems that will detect viruses in additional, potentially relevant species, and in both sick and healthy animals (the latter of which may be infected but asymptomatic). Once identified, decisions should be made to handle them suitably, and beyond the reasons of import/export. Von Dobschuetz et al. (2015) concluded that targeting more viruses, particularly subtypes with zoonotic risk, and animal species, routine testing, genetic sequencing, and reporting and granting access to results can help CVOs in their decision-making responsibilities. This highlights the importance of including experts in various fields in the team (as CVOs in the current study did), and implementation of practices (e.g., effective surveillance systems) to provide support in decision making.

4.3. The political economy of decision making

While CVOs emphasized science-based decisions, our results also highlight the influence of economic and political forces—what can be described as the political economy of veterinary public health. Disease events and control measures redistribute economic costs and benefits, often triggering responses from industry stakeholders and political actors (Zinsstag et al., 2011; Grace, 2015). Several CVOs noted lobbying pressures, the importance of stakeholder communication, and how public or political acceptance could shape not only what decisions were made, but when and how they were announced. This dynamic reinforces the idea that public health decisions are not made in a vacuum but within a broader system of institutional incentives and economic interests, a pattern observed in both historical (Gonçalves, 2000a) and contemporary crises.

While legal frameworks and protocols were commonly cited as guides for action, our findings suggest that these are not always sufficient in conditions of uncertainty. In some cases, regulations provided structure and legitimacy; in others, they were perceived as politically driven or misaligned with scientific best practices. This reflects broader debates in public health governance around the precautionary principle and the tension between regulatory rigidity and adaptive response (Hood, Rothstein, 2001; Vogel, 2012). Cultural and institutional differences—such as between European precautionary approaches and American cost-benefit models—may further shape how decisions are framed and justified across jurisdictions.

4.4. Trust, communication, and public accountability

Another major theme was the role of communication and public trust in legitimizing decision making. CVOs underscored the importance of transparency, humility (e.g., admitting uncertainty), and ethical clarity—values that help maintain public confidence during crises. These practices resonate with literature on trust in risk communication (O'Neill, 2002) and the importance of integrating public values into scientific decision making. The finding that “being scientifically correct” can conflict with public perceptions or cultural norms also echoes Oreskes’ (2010) work on the social construction of doubt and the political shaping of scientific narratives.

Collecting data on people’s views, particularly when the subject matter is sensitive, can be challenging. Indeed, in this study it proved somewhat arduous to recruit interviewees despite assurances of confidentiality, as many were uncomfortable in speaking freely. When conducting interviews, it is crucial to keep in mind that some terms, even though they may appear to be interchangeable, have subtle nuances in meaning, particularly across languages. Chan and Enticott (2023) discovered this in interviewing Chinese pig farmers on their perception of what a “good farmer” was. Rather than “good”, the term “diligent” was used by many to express their views; the distinctions have different cultural meanings. “Diligence” also involved aspects of making money, and participating in the life of the village, i.e. beyond the farm. In addition to the verbal interviews, the participants also sketched out

graphically maps of their farms (a narrated mapping approach), which helped them further convey their thoughts on “good” vs. “diligent” farming (Chan and Enticott, 2023). This may also have been the case in the present study, e.g. concerning the interview question “Have you had to make decisions that may or had compromised your career?”. While most interviewees appeared to view the word “compromised” in its negative connotation (which was the original intention), some viewed this term in a more positive light. That is, they interpreted “compromising” more as “negotiating” or “coming to a mutual agreement” with stakeholders and others.

4.5. Implications for policy and education

This study highlights the importance of decision preparedness, including the development of institutional memory, simulation exercises, and inter-professional networks, particularly in contexts when decisions are not clear-cut and different stakeholder interests are being balanced. It also suggests the need for governance frameworks that allow both scientific rigor and procedural flexibility, enabling CVOs to act swiftly while maintaining transparency and legitimacy. The findings underscore the value of interdisciplinary education for future veterinary leaders, who must integrate epidemiology, economics, ethics, and communication skills in their decision making.

Presently, in general, veterinarians do not receive adequate training in how to integrate evidence and other factors in order to make an informed decision. Therefore, veterinary education should include units on decision making and policy formulation. Such training would be relevant not just for future CVOs, but for all veterinarians in general. Workshops on these topics for veterinary students as well as established veterinarians, particularly those involved in policy development, would be of great value. The fact that some decisions are not straightforward and often require trade-offs is ample justification for additional training in ethics also, e.g., balancing costs and benefits for different groups, and developing frameworks for how to do that.

During the decision-making process, McKay (2011) recommends the use of appropriate tools, respecting the views of stakeholders, considering the pros and cons of all possible options, choosing the best one, informing all stakeholders, and then taking appropriate implementation steps. Such tools could be introduced during veterinary training.

4.6. Qualifications of a “good” CVO

Perhaps the major message arising from this work is that although all CVOs follow the established protocols in the end, the individual CVO’s “subjective” decisions play a surprisingly large role in making a “good” decision for the country and for the CVO’s own well-being in the position. The CVOs provided ample responses to support “subjectivity” in decision making (e.g., insufficient information and conflict between scientific evidence and political/economic pressures, i.e., the nature of the work; and personal characteristics of the CVO). Although being a CVO is stressful and not for everyone (e.g., “You shouldn’t be too nervous. It’s good if you’re an easygoing person, but not too easygoing. So a kind of a compromise”; and “It’s hard to make a decision: if it’s wrong, the impact will be huge.”; both CVOs with a high level of decision-making authority), most CVOs expressed a strong commitment to their work and felt they were making valuable contributions (e.g., “When we have problems in more than one region I must coordinate and try to have a common response in the different regions. It’s not easy, but it’s important work.”; and “We as vets work with people too. We’re here to help people manage their resources, their livestock, to get the best out of their livestock.” (also both CVOs with a high level of authority)). Therefore, the qualifications of the CVO are not limited only to subject matter knowledge but also include the personal qualities of the individual, or their “fit”. Both aspects should be considered in CVO appointments, and indeed, for all higher positions. Thus, to some extent, at least subject matter knowledge and the role of the CVO in decision making should be included in our education system and

beyond that, perhaps psychological testing, should be considered in the selection process. Given that very few veterinary students will actually become CVOs, it may not be practical to include such aspects in core veterinary curricula. Instead, it may be feasible to offer such training and focus on the role of a CVO as an elective subject, for students who already express an interest in becoming CVOs later in their career. Of course, a good, general understanding of veterinary services and veterinary administration is important for practitioners at any level of authority. Furthermore, it would seem worth training veterinary students in relationship building, leadership styles, and risk assessment / risk management more generally, even if they do not intend to become CVOs. Indeed, this seems highly relevant to any field making evidence-based decisions, and given the engagement that veterinarians have with the public, is not unique to the CVO arena (but would certainly be useful for those following the CVO path).

4.7. Limitations, strengths, and future research

This study is based on retrospective accounts and self-reported experiences, which may be shaped by recall bias or social desirability. Those that participated self-selected to join the study and share their experiences. Another limitation was that although the CVOs represented all 5 WOA regions, it was more difficult to find interviewees from 2 regions, Africa and Middle East, so there were some regional biases. Nevertheless, the experiences and responses of our participants from those regions did not markedly differ from those of other regions, so we do not expect that additional individuals from these regions would dramatically alter the results.

Some transcriptions from the interviews were difficult to interpret. However, most such instances were clarified by the context and did not detract from the overall understanding of the specific text.

Finally, the extent to which the actual role/position of an individual CVO impacts their views on the decision making process may be unclear. However, VAs are represented by WOA delegates, who in most instances are also the national CVO. WOA delegates are formally appointed by the Ministry responsible for livestock, e.g., the Ministry of Agriculture. Thus, it is assumed that the government has delegated decisions on animal health issues to the WOA delegate.

Despite these limitations, the study also has strengths. One strength of this study is that our interviewees came from many different countries in WOA's 5 operating regions (Africa, Americas, Asia and Pacific, Europe, and Middle East), which increased diversity, comparability, and generalizability of our findings. Our use of snowball sampling helped to identify some potential interviewees, and is a recommended approach for identifying key informants that are hard-to-reach (Bernard, 2017). This method has been used previously in various qualitative veterinary studies, as well. One such example involved communication preferences between clients and veterinarians during decision making (Groves et al., 2024). Another studied the sources of stress experienced by Hong Kong small animal veterinarians and how these could be alleviated (Chan and Wong, 2023). Toghroli et al. (2024) investigated obstacles that veterinarians face in dealing with antimicrobial resistance. Thus, snowball sampling has been used in a wide range of studies, particularly when trying to understand decision making in veterinary contexts.

Another strength is that, to our knowledge, no such study has previously been conducted. Finally, the results can be used to improve the education of veterinary students, by including modules on decision making in curricula, and offering workshops on decision making for professionals involved in animal health.

The consistency across interviews and diversity of contexts (e.g., across countries and disease types) enhances the robustness of our findings. Future research could complement this work with ethnographic or real-time studies of decision making during outbreaks, as well as comparative analysis of how institutional design influences the use of evidence under pressure.

As any public policies, veterinary public health interventions are not

implemented in a vacuum. We chose to interview CVOs as they represent their countries at the WOA, and are appointed by the Ministry of Agriculture. In addition, CVOs hold by nature a veterinary background which enables them to understand the biological nature of the challenge. In line with Kingdon (1984), we acknowledge that policy formation follows nonlinear and complex pathways involving multiple stakeholders advancing problems, policies and politics. In the public health field, policy networks studies emphasize the importance of the structures and the positions of stakeholders in the network (Johnson et al., 2012). In our study we did not focus on a specific policy formation, hence did not mobilize such theoretical frameworks. Yet, interviewing CVOs enabled us to capture the main determinants of decision-making of VAs, while recognizing the importance of other lower ranked government employees, surrounding the CVOs in their daily work.

Throughout the interviews, CVOs did not discuss specifically / distinguish risk assessment / risk management, and we did not specifically probe for this in the interview guide. This could be an area of potential future research, particularly given its relevance for trainees and veterinary students interested in these careers.

5. Conclusions

The aim of this study was to investigate the personal experiences of CVOs with challenging decision-making processes, particularly for urgent public health matters, and how this has led to changes in policies and practices. The data were collected by interviewing CVOs from all over the world and the responses were analyzed using qualitative methods. Echoing prior scholarship, our results illustrate the critical role of and challenges for CVOs during decision making when they have to integrate evidence from a variety of sources, managing risks while taking uncertainty and other factors, including economics and politics, into account, and maintaining trust, while being under pressure due to urgency and/or stakeholder interests.

Most CVOs followed established protocols and legislation, and relied on a multidisciplinary team of experts, including those with scientific, economic, policy, and legal expertise, and those with knowledge of local field conditions, for guidance. The lack of sufficient information and conflict between science and socio-economics and politics were common themes in many of the situations the CVOs discussed. Although stressful, most CVOs loved their job and felt they were making a valuable contribution towards animal health, and, directly or indirectly, also to human health. It would be beneficial for educational programs aimed at veterinary undergraduates as well as veterinary professionals to include components that will raise awareness about the complexity of veterinary decision making during animal and human health emergencies.

CRedit authorship contribution statement

Grohn Yrjo: Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Hertl Julia:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Data curation. **Elizabeth L. Fox:** Writing – review & editing, Investigation, Formal analysis. **Gregorio Torres:** Writing – review & editing, Investigation, Conceptualization. **Dirk Pfeiffer:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Guillaume Lhermie:** Writing – review & editing, Writing – original draft, Investigation, Data curation, Conceptualization.

Declaration of Competing Interest

With this statement all authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Invitation Letter (E-mail)

RE: How are veterinary public health decisions made?

Dear Dr. XXXX,

We are conducting a research study on the role of the veterinarian in multi-stakeholder decision making. Our focus will be on decisions requiring immediate action, which may have a national impact on animal and/or human health. Choosing the best option among several alternatives among competing interests is not straightforward. The criteria regarding how the decisions are made require considering not only animal and human health but socioeconomic factors, as well. Furthermore, risk assessment is an essential component of the decision-making process. This research project is affiliated with a university. Our particular interests are to investigate 1) how veterinarians make decisions requiring immediate action, which may have a national impact on animal and/or human health and 2) how the decision-making process creates new policies and practices. We would like to gather as much information as possible about this, which is why we are reaching out to you and other chief veterinary officers (both currently working and retirees). To the best of our knowledge, such data have never before been collected and would therefore be a novel contribution to the literature. Furthermore, presently veterinarians do not receive adequate training in how to integrate evidence and other factors in order to make a 'good' decision. We believe that our findings will therefore provide education value both to students and established veterinarians. If you are willing, we would like to invite you to participate in our study, which would involve interviewing you via Zoom, at a time and place convenient to you. The interview should last approximately one hour. As an attachment, you will find a consent form for this study. If you would like to participate, please complete and return the consent form to us. If you have any questions or concerns, please feel free to contact us. Then, if you would still like to participate, prior to the start of the interview, we will also need to obtain verbal consent from you. Thank you for your time, and we hope to speak with you soon!

Sincerely,

Author 1

Author 2

Author 3

Appendix B. Consent Form

We are asking you to participate in a research study titled "How are veterinary public health decisions made?" We will describe this study to you and answer any of your questions. This study is being led by Author 1.

What the study is about

The purpose of this research work is not to find the optimal way to make decisions but rather to investigate the role of the veterinarian in multi-stakeholder decision making. The focus will be on decisions requiring immediate action at the national level. The specific objective is to consider what is the role of Chief Veterinary Officers (CVOS) in the decision-making process for the control of animal diseases in situations, which may have a national impact on animal and/or human health.

What we will ask you to do

It is your choice whether you agree to participate. If you do agree, we will ask you to participate in either a face-to-face interview or a Zoom

interview at a time and location that is convenient for you. The interview will take approximately 60 min. We will ask questions about decisions you have made in acute situations, and your attitude and perceptions about the decision-making process and its potential impact on current and/or future policy and practice. Risks and discomforts We do not anticipate any risks from participating in this research. However, you may be uncomfortable answering some of the questions in the interview. However, you do not have to answer all the questions and may stop at any time. You just need to let us know.

Benefits

Although this research may not have any direct personal benefit to you, you may appreciate the opportunity to describe your experiences. The findings from the study will be published in a research paper, thus contributing to the literature. Furthermore, the findings will also be integrated into veterinary curricula in order to improve teaching of the decision-making process, and may also be the subject of training workshops.

Incentives for participation

Participants will not receive any incentives.

Audio/Video Recording

If you are OK with it, we would also like to record the interview, so that we do not have to take notes while we are speaking with you. We will not include any identifying information in the recording, and will delete the identifying information after we have reviewed it for our notes, and transcribed it. If you feel uncomfortable with the recording, you can still participate in the interview with us.

Verbal Consent

Would you be OK with us recording the interview?

Privacy/Confidentiality/Data Security

There is a risk that someone outside the study could see your information. However, we will do our best to keep any identifying information safe by replacing your personal information with a special code/identifier. Any materials with identifying information will be locked in a drawer and/or will be stored on Cornell's cloud 2 storage, which is encrypted, and which requires a password to access. All materials with identifying information will be shredded or deleted once the study is complete. As we have coordinated with you via email to coordinate interview time, please note that email communication is neither private nor secure. Although we are taking precautions to protect your privacy and are using Cornell's secure email platform, you should be aware that information sent through email could be read by a third party. We anticipate that your participation in this survey presents no greater risk than everyday use of the Internet.

Sharing De-identified Data Collected in this Research

De-identified data from this study may be shared with the research community at large to advance science and health. We will remove or code any personal information that could identify you before files are shared with other researchers to ensure that, by current scientific standards and known methods, no one will be able to identify you from the information we share. Despite these measures, we cannot guarantee anonymity of your personal data. Taking part is voluntary. Your participation in this research is voluntary. You may decide not to participate before we start the interview, stop the interview at any time, and/or skip any questions that make you feel uncomfortable.

Withdrawing will not impact your relationship with any organizations involved in this research project.

If you have questions

The main researcher conducting this study is Author 1. Please ask any questions you have now. If you have questions later, you may contact Author 1 by email or phone. If you have any questions or concerns regarding your rights as a subject in this study, you may contact the Institutional Review Board (IRB) at our institution. You may also report your concerns or complaints anonymously through Ethicspoint online or by calling toll free at 1-866-293-3077. Ethicspoint is an independent organization that serves as a liaison between the institution and the person bringing the complaint so that anonymity can be ensured.

Statement of Verbal Consent

I have read the above information, and have received answers to any questions I asked. I give my verbal consent to take part in the study.

Appendix C. The actual interview questions, including probes

Interview Guide

A. INTRODUCTION

Thank you for agreeing to take part in this study. As we described before, the purpose of this study is to learn about your views and experiences on the decision-making process and its influence on policy and practice in veterinary medicine and public health.

Do you give your verbal consent to participate in and record this interview?

Before we start, I would like to remind you that there are no right or wrong answers. We are interested in hearing your story and knowing what you think, so please feel free to share your point of view and opinion.

[If participant agreed to recording]: May I begin the audio/Zoom recording?

NOTE for interviewer: If participant agreed to being recorded, turn on tape recorder/check in Zoom to verify that it is working. Otherwise take as detailed notes as possible.

NOTE for interviewer: The following is a guide. Potential probes are included, but they are only intended to support the main question and the flow of the discussion is the priority. The intention is not to read through all of the probes for each question.

1. Background, setting

1.A. First, I want to hear about your experiences in veterinary medicine. What sort of veterinary work did you do prior to becoming a veterinary officer? [record whether current or ex-officer] Tell me more about that.

(If not mentioned in response) 1.A.1 What is your total length of experience in veterinary medicine, not including vet school? [

(If not mentioned in response) 1.A.2

PROBE: What sort of practice did you work in? What sort of teams did you work with, if any?

PROBE: How did those experiences inform the work you do/did as a veterinary officer, if at all?

1.B. Transitioning now to your role as a veterinary officer, how did you come to become a veterinary officer [record whether current or ex-officer]?

2. Organization and responsibilities

2 A. What geographical area falls/fell under your jurisdiction as chief veterinary officer? How did you get involved in that region/position?

(If not mentioned in response) 2 A.1. Who do you usually work closest with? Tell me about those interactions.

(If not mentioned in response) 2 A.2. Who do you report to? What does that reporting process look like?

2.B. To what extent does your position empower your decision making?

2.C. Could you describe the architecture of your organization/ministry organizations?

(If not mentioned in response) 2C1. Could you tell me about the coordination between the ministries?

PROBE: Ask for information about the governance structure (who reports to whom; maybe ask for a “chain of command”): The set up in different jurisdictions would affect the decisions that CVOs are allowed to make.

3. Others involved, timing

3.A. If an urgent situation arises in your jurisdiction in which a quick decision needs to be made, how is that decision typically made?

PROBE: who makes the decision? Do you make it alone, or in consultation with others? In what ways are different people involved (or not)?

PROBE: Specify exactly who these others are: where they are located (same or different jurisdiction); are they local vets “on the ground”, other CVOs in other areas, subject-matter experts?

3.B. Tell me about one of these urgent situations that you’ve been involved with in your role as a veterinary officer.

PROBE: How much time elapsed between when you first learned that a decision needed to be made, and when it was actually made? What influenced that timeframe?

(if participant mentions slow decision-making process) PROBE: What were some of the reasons that the decisions were delayed?

(if participant mentions quick decision-making process) PROBE: What were some of the things that facilitated a quick response in this situation?

PROBE: Did the length of time cause you any concern? Tell me more about that.

4. Sources of information for “typical” and “unusual” situations

4.A. Now I would like to discuss decision making surrounding both a typical situation, and then a very unusual one.

First, let’s start with a typical decision that you might make in your role as a veterinary officer. What is a typical decision that you might make in your role as a veterinary officer?

Who did you think to contact first for more assistance or information?

4.B. What information did you use to guide you in making the decision?

PROBE: Past personal experience with similar problems; colleagues’ experience; common cultural practices (e.g. respecting India’s reverence for cows); Internet search; action plans, regulations, the literature and/or the specific research done during the situation, and expert opinions, including mathematical models done for different scenarios. [ASK THEM TO DISCUSS IN DETAIL EACH SOURCE THAT THEY USED.]

4.C. Of all of the sources of information that we have just discussed, which was the most influential in guiding you through the decision-making process? How influential was each source? Which was least informative?

PROBE: What information was missing, which would have been helpful to you? E.g. number of cases in the past week (or whatever time period); where/when were the cases occurring (and was there any spatial or temporal pattern?); what treatments/ interventions have already been tried, and by whom; could the source have been contaminated feed, water etc., or a natural event (e.g. catastrophic flooding or hurricane etc.)? How do you evaluate and validate its value?

4.D. Now I would like to hear about an unusual decision you had

to make, that was very challenging or that has stuck with you. **Tell me about that decision and what made it unusual.**

Who did you think to contact first for more assistance or information?

4.E. What information did you use to guide you in making the decision?

PROBE: Past personal experience with similar problems; colleagues' experience; common cultural practices (e.g. respecting India's reverence for cows); Internet search; action plans, regulations, the literature and/or the specific research done during the situation, and expert opinions, including mathematical models done for different scenarios. [ASK THEM TO DISCUSS IN DETAIL EACH SOURCE THAT THEY USED.] On which past experience? On which tool do you rely? How much do you rely on the expertise of the private sector? Could you give examples?

4.F. In situations of high uncertainty, how do you deal with the pressure of quick answering?

4.G. Of all of the sources of information that we have just discussed, which was the most influential in guiding you through the decision-making process? How influential was each source? Which was least informative?

PROBE: What information was missing, which would have been helpful to you? E.g. number of cases in the past week (or whatever time period); where/when were the cases occurring (and was there any spatial or temporal pattern?); what treatments/ interventions have already been tried, and by whom; could the source have been contaminated feed, water etc., or a natural event (e.g. catastrophic flooding or hurricane etc.)?

4.H. What information do you wish you had when you were making that decision? Why is that?

PROBE: Can you name a specific veterinary crisis that was particularly difficult to handle? Was there a situation that was particularly easy?

4.I. If you have never faced such a situation before, or have only limited knowledge about it, how did you proceed? I.e., did you make a decision anyway, or did you wait until you could learn more about it?

5. Other influences

Now I'd like to discuss other factors that may have influenced your decision making, and compare and contrast how those might differ between the typical situation you described, and the "unusual" one. Please emphasize how these differ between the typical and unusual situation.

For unusual situation:

5.A. What alternative decisions or choices were available to you?

5.B. How did time limitations or competing priorities or deadlines imposed on you force you to act?

PROBES: How often do you have to deal with multiple challenges simultaneously? How do you prioritize?

5.C. Were financial (budget) or other resources constraints (e.g. lack of diagnostic equipment or tests) a factor in your decision-making process? If yes, could we discuss these in more detail?

PROBE: Are there some organizational constraints limiting your action?

5.D. Can you describe interactions in the decision-making process?

5.E. This next question is a sensitive one, but as we have assured you previously, your responses will remain confidential, so you may feel free to discuss such concerns here. How did politics or power influence the decision?

PROBE: How would the decision have been different without concerns about politics or power?

5.F. In what ways did public acceptance impact the decision?

PROBE: Did public acceptance influence you during the process? If

so, how?

PROBE: Did you do anything differently because of public acceptance? Tell me more about that.

5.G. Did you observe administrators guided by selfish interest and not social interests?

5.H. Have you had to make decisions that may or had compromised your career?

5.I. Do any other things come to your mind that may have influenced your decision making, besides the ones we have just discussed?

6. Things learned and/or implemented (for both the "typical" and "unusual" situation)

Consequences of decision

6.A. Did you form a follow-up plan to modify the decision, if need be?

PROBE: If so, what was it? If not, why not?

6.B. Did your decision change or influence policy or practice, or create a new policy or practice?

What was learned (for unusual situation only)

6.C. Has the process led to policy and/or practice change? If yes, let's discuss what the changes have been. Or if not, why do you think that no changes have occurred?

6.D. What did you learn as a result of going through this decision-making process, for the unusual situation that you faced?

PROBE: Did you learn more about yourself, or about other people involved? If so, what did you learn? Did you learn more about the actual problem, or how stakeholders work together (or not, as the case may be)? Were there conflicts among the people you consulted (e.g. stepping on other people's toes, or overstepping one's authority if there are disagreements)?

6.E. When it comes to decision making, what do you wish you had known or been trained on during your training?

6.F. What has most surprised you about the decision-making process?

6.G. From your experiences, what is most challenging about the decision-making process?

7. Open remarks 7.A. Is there anything related to the purpose of this meeting you wish to add?

Thank you for your time! May we contact you later on if we need clarification regarding any of your responses?

If you are interested, we will provide you with a summary of our findings when we have compiled them. No identifying information will be shared, and your responses will remain completely confidential.

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